

PROJECT QUANT — GVM DEVELOPMENT MEMORY

G-Series | Build Journey G1 to G8 | Master Sequence Locked

Arpit Goel + Claude | April-May 2026

CORE MISSION

93% of retail investors in India lose money — not because they lack intelligence, but because they lack the tools institutions have. Project Quant exists to change that.

Build AI Aladdin — India's first institutional-grade stock intelligence platform for retail investors. A platform where a 25-year-old investor in Lucknow has the same analytical edge as a fund manager in Mumbai.

BUILD STATUS — G1 TO G8

Session	Date	Key Achievement	Status
G1	Apr 2026	Project vision locked. Tech stack decided. Python + VS Code install planned.	Done
G2	Apr 2026	Python 3.14 + VS Code + Pandas installed. gvm_scorer.py created.	Done
G3	Apr 2026	gvm_engine.py built. screener_loader.py reading CSV. All 21 parameters mapped.	Done
G4	30 Apr 2026	6 logic fixes applied. 1,806 stocks scored. 127 segments target set.	Done
G5	7 May 2026	3,371 merge issue found. PostgreSQL setup next. Pending duplicate fix + validation.	Done
G6	19 May 2026	Platform architecture expanded. Internal EXE track added. API Library defined.	Done
G7	20 May 2026	Railway-first architecture locked. No MCP. No Cowork. No Desktop dependency.	Done
G8	21 May 2026	Input 2 cleaned. 126 segments locked. Analytics layer merged. GitHub live.	Done

G8 — TODAY'S ACHIEVEMENTS (21 May 2026)

Posture	Task	Result
1	Code fixes — screener_loader merge bug, segments, Stylam	Done — 1,757 stocks, 126 segments
2	5-stock validation gate	Done — pre-validated in earlier session
3	GitHub setup + first commit + push	Done — github.com/AGQuant/quant_project (Private)
4	PostgreSQL on Railway	Pending — next session
5	FastAPI deploy on Railway	Pending — next session

Key Decisions Locked in G8

- **Input 2 = Master Universe rule** — Final universe always defined by Input 2 NSE codes. Screener.csv role limited to providing parameter values. Stocks without screener match are excluded from DB entirely (no default-5 padding, no BSE fallback).
- **Database principle locked** — DB = truth. Every row in gvm_scores table must be real data, real score, real rating. No synthetic neutrals.
- **Analytics layer merged** — commentary_generator.py absorbed into gvm_analytics.py (single script, single output: gvm_analytics.csv). gvm_scorer.py and commentary_generator.py deprecated.
- **Segment count finalized at 126** — 7 small segments merged into parents (Chemical Smallcap, Engineering Smallcap, FMCG Foods Small, Hotels & Hospitality, Trading, IT Smallcap, Electronics Small). KN Agri, PFC, Stylam manually re-segmented at source.
- **GitHub repo live** — github.com/AGQuant/quant_project. Private. 17 files committed. .gitignore protecting service_account.json. Memory PDFs included as backup.

TECH STACK — LOCKED

Component	Technology	Status
Backend	Python + FastAPI	Built (main.py — 25 endpoints)
Frontend	React	Week 3
Database	PostgreSQL on Railway	G9 next task
Hosting Backend	Railway only — no AWS	Pending deploy
Hosting Frontend	Vercel (free tier)	Pending deploy
Data Source	Screener.in CSV + Input 2 sheet	Live
Historical + Price Data	Yahoo Finance (yahoo_hist.py + yahoo_quote.py)	Live — no token needed
AI Layer	Claude API	Week 4
Version Control	GitHub — github.com/AGQuant/quant_project	Live (G8)
Sheet Reading	gsread + service account	Installed
MCP	REMOVED — not needed	Deleted
Fyers API	REMOVED permanently	Deleted
AWS	NOT USED — Railway only through Year 1	Decided

RAILWAY-FIRST ARCHITECTURE — LOCKED

Decision locked 20 May 2026: No MCP. No Cowork. No Desktop dependency. No AWS. Railway is the brain. Browser is the interface.

Layer	What it does	Status
PostgreSQL on Railway	Stores all GVM scores, price history, OHLC, signals	G9 next task
FastAPI on Railway	Exposes all data as API endpoints — always on	Week 2
Browser Claude	Fetches Railway URLs — reads live data anywhere	After Railway live
VS Code	Write and deploy code only — no data dependency	Active
Google Sheets	Source data via gsread — live prices only	Active
GitHub	Version control — github.com/AGQuant/quant_project	Live

2 MODELS ONLY — FINAL DECISION

Pivot decision May 20, 2026: Build 2 models via chat only. Quality over speed. Depth over breadth.

#	Model	Why Priority	Timeline
1	GVM Rating Engine	Core product — 85% built post G8	Month 1-2
2	V5 Signal Engine	Live trading — real P&L impact daily	Month 2-4

All other models (Portfolio Health, Screener, Alerts, Pair Trading) — parked indefinitely.

MASTER BUILD SEQUENCE — UPDATED POST G8

Phase 1 — Foundation (G8 COMPLETED)

#	Task	File	Status
1	Fix duplicate merge	screener_loader.py	■ Done
2	Fix segment count to 126	input.csv + loader	■ Done
3	Fix Stylam, PFC, KN Agri segments	input.csv	■ Done
4	Validate 5 stocks	gvm_engine.py	■ Done (pre-validated)
5	Merge analytics + commentary	gvm_analytics.py	■ Done
6	GitHub setup + first commit	Repo: AGQuant/quant_project	■ Done

Phase 2 — Database & Deploy (G9 NEXT)

#	Task	Detail
7	PostgreSQL on Railway	quant_db live. 4 tables: gvm_scores, raw_prices, signals, users
8	Insert Day 1 gvm_analytics.csv	1,757 stocks scored — first day stored
9	FastAPI deploy on Railway	main.py live with all 25 endpoints
10	Browser end-to-end test	Claude fetches Railway URL — confirms

Phase 3 — Raw Data Layer (Week 2)

#	Task	Detail
11	Download 5Y OHLC for 1,971 stocks + Nifty50 + Bank Nifty	Back Nifty via yahoo_hist.py
12	Store all OHLC in Railway PostgreSQL	raw_prices table
13	Daily OHLC update scheduler	3:45 PM IST, Mon-Fri only
14	Momentum parameters via FastAPI	1Y/3Y Return, DMA50, DMA200, 52W vs Index
15	Daily GVM score regeneration	All 1,757 stocks rescored daily, date-stamped

Phase 4 — Frontend (Week 3)

React on Vercel. Search bar → stock name → GVM rating + breakdown + commentary punchline. Backend hits Railway FastAPI.

Phase 5 — AI Layer (Week 4)

Claude API integrated for stock explanation. Plain English commentary on top of GVM scores.

GVM MODEL ARCHITECTURE — 21 PARAMETERS

G — Growth (14 Parameters)

Operational (9): 5Y Sales Growth, 3Y Sales Growth, 5Y Profit Growth, 3Y Profit Growth, QoQ Sales Growth, QoQ Profit Growth, OPM, OPM Expansion, Fixed Asset Growth

Reliability (5): Promoter Holding, Institutional Holding Change, ROCE, Interest Coverage (skip BFSI), Dividend Yield

V — Value (2 Parameters)

PE vs Peer Group (inverted logic — lower PE = higher score) | Potential Annual Upside

M — Momentum (5 Parameters)

1Y Return, 3Y Return, DMA50, DMA200, 52-Week vs Index — ALL computed from Yahoo raw OHLC stored in Railway

Scoring Logic

2-Factor System: Factor 1 (absolute performance) + Factor 2 (relative vs peer group average), averaged. Scale: 2.5 / 5 / 7.5 / 10. Absolute thresholds: >15% = 10, 5-15% = 7.5, 0-5% = 5, below 0% = 2.5. PE uses inverted logic. Peer averages filtered to exclude outliers (10-90 percentile trim). GVM = G + V + M averaged.

ANALYTICS LAYER — gvm_analytics.py (G8 NEW)

Single script generates scoring + commentary in one pass. Replaces gvm_scorer.py and commentary_generator.py.

Score Band	Label	Verdict
≥ 8.0	Excellent	Strong Buy
7.0 – 8.0	Good	Buy
6.0 – 7.0	Average	Accumulate
5.0 – 6.0	Below Average	Wait & Watch
< 5.0	Poor	Avoid

Day 1 Verdict Distribution (1,757 stocks)

Verdict	Count	% of Universe
Strong Buy	69	3.9%
Buy	390	22.2%
Accumulate	608	34.6%
Wait & Watch	532	30.3%
Avoid	158	9.0%

QUANT_PROJECT FOLDER — POST G8 STATE

File	Purpose	Status
gvm_engine.py	Core scoring logic — all 21 parameters	v3 — D5 fixes applied
screener_loader.py	Reads CSV, cleans, merges, calls engine	v3 — 126 segments stable
gvm_analytics.py	Scoring + commentary in one pass	NEW — G8 deliverable
input.csv	Master universe — 1,971 stocks, 126 final segments	G8 corrected
screener.csv	Screener.in export — 1,922 stocks (Above 500 Cr query)	Latest
main.py	FastAPI app — 25 endpoints	Built — pending Railway deploy
yahoo_hist.py	Yahoo historical OHLC	Live — no token
yahoo_quote.py	Yahoo live quote (15min delay)	Live
yahoo_mcp.js	Legacy MCP — kept as archive	Archived
Procfile	Railway deploy config	Ready
requirements.txt	Python dependencies	Live
service_account.json	Google Sheets auth	Local only (.gitignored)
.gitignore	Git exclusions	G8 created
gvm_scorer.py	Legacy v1 scorer	DELETED G8
commentary_generator.py	Legacy commentary script	DELETED G8
gvm_results.csv	Legacy output	DELETED G8
gvm_commentary.csv	Legacy output	DELETED G8
package.json + lock + node_modules	MCP leftovers	DELETED G8

GROUND RULES — NON-NEGOTIABLE

- **Input 2 = Master Universe.** 1,971 NSE codes. Output = 1,757 (only stocks with screener match). 214 unmatched stocks ignored from DB entirely.
- **No default-5 padding.** If we don't have data, we don't fake a rating. DB = truth.
- **No PostgreSQL until validation passes.** Validated — moving to G9 DB setup.
- **Railway-first, Railway-only.** No AWS through Year 1.
- **No MCP ever again.** FastAPI on Railway replaces MCP permanently.
- **No Fyers ever again.** Yahoo Finance handles all price data — free, no token.
- **Momentum from Yahoo raw data.** DMA50, DMA200, returns — all computed from stored OHLC.
- **Daily OHLC update.** Post-market scheduler — 3:45 PM IST, Mon-Fri only.
- **Blank values = 5.** Missing data defaults to neutral score (engine logic, not padding).
- **Interest Coverage — skip BFSI.** Banks/NBFCs excluded from this parameter.
- **2 models only.** GVM + V5. All others parked indefinitely.
- **service_account.json never enters GitHub.** .gitignore protects it.
- **Learning rule.** Explain WHY before HOW. Arpit directs. Claude executes.

DATA SOURCE RULE — NON-NEGOTIABLE

Yahoo Finance (yahoo_quote.py + yahoo_hist.py) = ALL price data. Historical daily and near-live quotes. No token. No auth. Always works.

Fyers = PERMANENTLY DELETED. Never reinstall. Never reference.

MCP = DROPPED. Railway FastAPI replaces MCP permanently.

AWS = NOT USED. Railway-only through Year 1.

G9 — NEXT SESSION COMMAND

*PostgreSQL on Railway → quant_db live → 4 tables created → Day 1 gvm_analytics.csv inserted → FastAPI deploy
→ Browser end-to-end fetch confirms. Ground rule: no new code on PC until Railway DB is live and inserting cleanly.*

G Memory — GVM Development | Updated 21 May 2026 (Post G8)